Arabic Sentence Segmentation Shared Task 2026

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Abstract

We present the findings of the Arabic Sentence Segmentation (**ARASEG**) Shared Task, organized at the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026). Given an Arabic document, the task requires systems to identify sentence boundaries and segment the text into coherent sentences. Systems are evaluated on the genre-diverse **ARASEG** corpus under four input settings that vary in punctuation availability and paragraph structure, each with a Closed Track restricted to the provided training data and an Open Track permitting publicly available external resources. 22 teams from 13 countries registered, 16 submitted systems, and 14 submitted system description papers. Across the eight subtask-track combinations, the best systems achieve F_1 scores ranging from 89.1 to 95.3. Performance declines as segmentation cues are removed, with the largest drop occurring when both punctuation and paragraph structure are unavailable. Pretrained-encoder ensembles and careful prediction calibration emerge as the most effective approaches, while gains from LLMs, external data, and complex architectures vary across systems and settings.[1]

1 Introduction

Sentence segmentation is a fundamental preprocessing step in natural language processing (NLP), as many datasets and downstream systems operate on sentence-delimited text. Sentence boundaries are important for tasks such as dependency parsing, named entity recognition, retrieval, and increasingly for large language model (LLM) pipelines, including text chunking, alignment (Qiu et al., 2025), and efficient training (Zheng et al., 2025).

Arabic presents unique challenges for sentence segmentation. Punctuation can be sparse, inconsistent, or absent altogether, especially in historical, literary, and other less standardized forms of text. Sentence boundaries may instead be signaled through discourse structure, coordinating conjunctions, and lexical cues, making them difficult to recover using punctuation alone. Even when punctuation is present, it does not always provide an unambiguous boundary signal (Yagi et al., 2024). These characteristics make Arabic sentence segmentation highly dependent on genre, document structure, and the availability of explicit surface cues, motivating evaluation settings that go beyond well-punctuated text.

To promote research on Arabic sentence segmentation, we organized the Arabic Sentence Segmentation (**ARASEG**) Shared Task as part of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026), co-located with EMNLP 2026. The shared task uses the **ARASEG** dataset, a manually annotated, genre-diverse corpus spanning eight genres with substantial variation in writing style, punctuation usage, and document structure.

The shared task consists of four subtasks defined by two dimensions: punctuation availability and paragraph structure. Figure 1 illustrates the differences between the four subtasks. Each subtask includes a **Closed Track**, where participants may use only the provided **ARASEG** data, and an **Open Track**, where publicly available external resources are permitted. These settings evaluate systems under increasingly challenging conditions, from text with explicit structural and punctuation cues to input where such cues are limited or absent.

A total of 22 teams from 13 countries registered for the shared task, with 14 teams submitting system description papers. Participating teams explored ensembles of pretrained Arabic encoders, task-adaptive training, calibrated thresholding, structured decoding, and LLM-based boundary refinement. This paper describes the task, participating systems, and official results, and analyzes the effects of input cues and modeling choices across settings.

[1]https://www.araseg.aramlab.ai/

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PA	أخبار	الفن.	¶	أعلنت	المنصة	عن	فيلمها	الجديد،	وسيعرض	قريبا.
NP	أخبار	الفن.		أعلنت	المنصة	عن	فيلمها	الجديد،	وسيعرض	قريبا.
NoPnx-PA	أخبار	الفن.	¶	أعلنت	المنصة	عن	فيلمها	الجديد.	وسيعرض	قريبا
NoPnx-NP	أخبار	الفن.		أعلنت	المنصة	عن	فيلمها	الجديد.	وسيعرض	قريبا

Figure 1: Four parallel variants of a document in the **ARASEG** corpus. The ¶ symbol marks paragraph boundaries, while red, boldfaced, and underlined words mark positions after which sentence boundaries occur.

2 Related Work

2.1 Sentence Segmentation

Sentence segmentation has been approached using a range of rule-based, statistical, and neural methods. Traditional systems rely on punctuation patterns and handcrafted rules, as in SpaCy$_{\text{SENT}}$ (Honnibal et al., 2020) and PySBD (Sadvilkar and Neumann, 2020). Other approaches infer boundaries from distributional patterns in unannotated text; Punkt (Kiss and Strunk, 2006), for example, learns abbreviation and sentence-boundary behavior without supervised training. More recent multilingual systems such as Ersatz, Where's the Point (WtP), and Segment Any Text (SaT) (Wicks and Post, 2021; Minixhofer et al., 2023; Frohmann et al., 2024) use pretrained transformer representations to predict sentence boundaries across languages. LLM-based segmentation has also received recent attention (Retkowski and Waibel, 2026), although specialized encoder models can remain competitive or superior in low-resource and historically diverse settings (Bilitski et al., 2026).

Arabic has received considerably less attention as a standalone sentence segmentation problem. Related work has investigated neighboring tasks, including clause segmentation (Ameur et al., 2008; Keskes et al., 2012), semantic text chunking (Alammar et al., 2025; Chirkunov et al., 2026), and punctuation restoration (Alshanqiti et al., 2022). Mekki et al. (2022) examined sentence segmentation for Tunisian Arabic, but broad evaluation across Arabic genres and document conditions remains limited. Although several multilingual segmentation systems include Arabic, such evaluations generally do not isolate the properties of Arabic text that make boundary detection difficult. The **ARASEG** Shared Task is designed to address this gap by providing a common evaluation setting in which systems are tested under different combinations of punctuation and paragraph information.

2.2 Arabic Sentence Segmentation Challenges

Arabic sentence segmentation is particularly challenging due to the ambiguous and inconsistent use of punctuation. Modern punctuation conventions entered Arabic writing relatively late, and many historical and literary sources contain sparse or no punctuation. In such texts, relationships between clauses and sentences are frequently expressed through conjunctions and discourse markers such as و *wa*[2] 'and', ف *fa* 'so/then', and ثم *θum∼a* 'then/afterwards'. As a result, identifying a sentence boundary may require lexical and contextual information rather than a dedicated boundary symbol. In **ARASEG**, non-punctuation-based boundaries account for 39% of sentence boundaries, varying across genres from 3% in Wikipedia to 59% in educational and 40% in religious texts.

Punctuation also provides an imperfect signal when it is present. Its use varies across genres, authors, and writing conventions, and the same mark may serve different discourse functions. The Arabic comma (،), for instance, can separate elements within a sentence or occur at a sentence boundary (Hadrich Belguith et al., 2005; Zaghouani and Awad, 2016; Yagi et al., 2024). Punctuation errors are also common in Arabic writing (Zaghouani et al., 2014), further reducing its reliability as a segmentation cue.

These properties directly motivate the design of the shared task (§3). By varying both punctuation availability and paragraph structure, the four subtasks measure the extent to which systems rely on explicit surface cues versus lexical and contextual evidence. The settings range from paragraph-aware, punctuated text to substantially harder input in which both punctuation and paragraph boundaries are removed, allowing us to assess robustness across genres and document conditions.

[2]Arabic HSB transliteration (Habash et al., 2007).

3 Shared Task Description

The goal of the **ARASEG** shared task is to segment Arabic documents into sentences. Given a document, systems must identify sentence boundaries to produce coherent sentence units. We formulate Arabic sentence segmentation as a binary token classification task, where the goal is to predict whether a sentence boundary follows each token.

The shared task is designed as a low-resource evaluation setting. Participants are provided with a fixed subset corresponding to **10%** of the available training data, sampled across genres, and use this subset for system development in the restricted setting. The task comprises four subtasks that vary along two dimensions: punctuation availability and paragraph structure. Each subtask is evaluated under two tracks.

Subtasks Participants compete in one or more of the following subtasks.

- **No-Punctuation No-Paragraph (NoPnx-NP)**: punctuation, domain-specific boundary markers (e.g., verse numbers in Quran/OT/NT and question/multiple-choice markers in ArabicMMLU), and paragraph boundaries are removed.
- **No-Punctuation Paragraph-Aware (NoPnx-PA)**: punctuation and domain-specific boundary markers are removed, while paragraph boundaries are retained.
- **No-Paragraph (NP)**: punctuation is retained, but paragraph boundaries are removed.
- **Paragraph-Aware (PA)**: punctuation and paragraph boundaries are retained.

Tracks Participants compete in one or more of the following tracks.

- **Closed Track**: Systems must be trained exclusively on the provided training subset, without using any additional training data.
- **Open Track**: Systems may use additional publicly available data for training.

With four subtasks and two tracks, the task results in a total of **eight** possible combinations. Participants are allowed to compete in multiple subtasks and tracks.

4 Shared Task Data and Evaluation

In this section, we present the datasets used in different tracks, describe the evaluation metrics, and outline the submission guidelines given to the participants.

Genre	Source	#Docs	#Paras	#Sents	#Tokens
Children	Green Library	11	200	534	8K
	Kashkul	1	29	30	185
	Majed	86	2K	3K	32K
	Mama Bread	1	23	38	437
	Spacetoon	12	283	283	1K
	chatGPT	2	248	248	2K
Education	ALC	5	24	213	3K
	ArabicMMLU	99	2K	9K	57K
	BTEC	6	584	584	5K
	Curriculum	39	3K	4K	40K
	ZAEBUC	39	70	433	6K
Literature	Arabian Nights	11	18	360	4K
	Hayy	9	166	525	9K
	Hindawi	67	1K	4K	59K
	Kalima	18	251	765	13K
	Sara	8	634	866	14K
Media	QALB	8	82	350	4K
	Subtitles	1	50	50	321
	WikiNews	28	156	365	6K
Poetry	Hanging Odes	5	329	329	3K
	Lang. Sings	3	75	75	404
	Al Issa	1	100	100	333
Politics	Constitutions	3	2K	3K	47K
	UN DEC HR	1	79	91	1K
Religion	Hadith	33	34	428	4K
	NT	28	191	1K	21K
	OT	29	196	996	18K
	Quran	63	63	2K	38K
Wiki	Wikipedia	41	345	1K	32K
Total		658	14K	36K	428K

Table 1: **ARASEG** shared-task statistics across genres and sources, including documents (Docs), paragraphs (Paras), sentences (Sents), and tokens (Tokens).

4.1 Data

Corpus Overview The shared task uses the Arabic Sentence Segmentation (**ARASEG**) corpus (Elkholy et al., 2026), a manually annotated, genre-diverse dataset developed for Arabic sentence segmentation. **ARASEG** consists of 2,187 documents containing 48,250 paragraphs and 121,623 sentences, totaling 1.4M tokens from 30 sources across eight genres. For the shared task, we adopt a low-resource setting and provide participants with a genre-stratified **10%** subset of the original **ARASEG** training (Train) split. The original development (Dev) and test (Test) splits are retained for model calibration and selection and for model evaluation, respectively. Table 1 summarizes the shared-task data across genres and sources.

Split	#Docs	#Paras	#Sents	#Tokens
Train	174	3,903	10,657	122K
Dev	222	5,066	12,985	156K
Test	262	5,025	12,509	150K
Total	658	13,994	36,151	428K
Blind Test	312	5,515	14,174	192K

Table 2: **ARASEG** shared-task split statistics by documents (Docs), paragraphs (Paras), sentences (Sents), and tokens.

Split	Pnx	Word	Overall
Train	6,034	4,623	10,657
Dev	8,198	4,787	12,985
Test	7,679	4,830	12,509
Total	21,911	14,240	36,151
Blind Test	10,795	3,379	14,174

Table 3: **ARASEG** shared-task sentence boundary statistics across punctuation-based (Pnx) and word-based (Word) boundaries.

Blind Test Set We introduce a new Blind Test set specifically created for the shared task and annotated following the same guidelines as **ARASEG**. The Blind Test set consists of 312 documents, 5,515 paragraphs, and 14,174 sentences. The Blind Test set is used for the official shared-task evaluation.

Corpus Variants **ARASEG** provides four corpus variants that share the same sentence-level annotations but differ in their input representation along two dimensions: punctuation availability and paragraph structure. These variants correspond to the four shared-task subtasks described in §3.

Table 2 presents the dataset statistics across splits. Additionally, Table 3 distinguishes between punctuation-based (Pnx) and word-based (Word) sentence boundaries.

4.2 Evaluation Metrics

We evaluate models using boundary-level precision (P), recall (R), and F_1, where predictions are correct only if they exactly match gold boundaries. We report macro-averaged document-level F_1 to avoid bias toward larger documents. We do not report P_k (Beeferman et al., 1999) or WinDiff (Pevzner and Hearst, 2002), which target coarse-grained segmentation rather than sentence boundary detection.

4.3 Submission Guidelines

The shared task is organized in two phases: development and testing. During the **development phase**, we set up CodaBench (Xu et al., 2022) competitions for all tracks. Participants may either evaluate their systems locally on the **ARASEG** Dev and Test sets,[3] or submit their predictions on the **ARASEG** Test set through the corresponding CodaBench competition for each track. Since the **ARASEG** Test set is publicly available, anyone is welcome to participate in this phase.

In the **testing phase**, we release the Official Blind Test set exclusively to registered participants. Registered teams are required to submit system description papers.

Teams are permitted to participate in all subtasks and tracks; however, their submissions must adhere to the resource constraints defined for each track.

5 Participants and Systems

We received 22 team registrations from 13 countries, with 16 teams submitting systems and 14 submitting system description papers. In total, we received 143 submissions during the development phase and 769 submissions during the testing phase. Table 4 lists the 14 teams that submitted system description papers along with their affiliations and the subtask–track combinations they joined.

5.1 Baseline

We establish a reference baseline by fine-tuning CAMeLBERT-MSA (Inoue et al., 2021), the best-performing model reported by Elkholy et al. (2026). We fine-tune it on the **ARASEG** shared task training data using the same training procedure and hyperparameters.

5.2 Summary of Submitted Systems

Table 5 summarizes participating teams' approaches and their rankings. Most systems relied on Arabic pretrained encoders, particularly CAMeLBERT (Inoue et al., 2021), AraBERT (Antoun et al., 2020), AraELECTRA (Antoun et al., 2021), ARBERT and MARBERT (Abdul-Mageed et al., 2021), and Arabic-BERT (Safaya et al., 2020). Several teams used multilingual or segmentation-specific models such as XLM-RoBERTa (Conneau et al., 2020) and SaT (Frohmann et al., 2024). Only four teams used

[3] https://github.com/mbzuai-nlp/araseg-shared-task-2026

Team	Country	Affiliation	PA		NP		NoPnx-PA		NoPnx-NP	
			C	O	C	O	C	O	C	O
Abjad AI	Saudi Arabia	Abjad AI	✓		✓					
Al-Qahera	Egypt	AUC	✓	✓	✓	✓	✓	✓	✓	✓
AUC DL Lab	Egypt	AUC	✓	✓	✓	✓	✓	✓	✓	✓
Axiom	Bangladesh	Chittagong University		✓						
DxMultiHead	Saudi Arabia	KFUPM	✓		✓		✓		✓	
ENSAR	Egypt	AUC	✓	✓	✓	✓	✓	✓	✓	✓
FAAMG	United States	AUC							✓	
iMak AI Lab	Russia	iMak AI Lab	✓		✓		✓		✓	
Itgan	United Kingdom	Independent	✓		✓		✓		✓	
Knights-ST	Egypt	Alexandria University	✓							
Mostafa ElAraby	Canada	Université de Montréal								✓
NAMAA-Community	Germany	NAMAA Community	✓	✓	✓	✓	✓	✓	✓	✓
NLPBeasts	Egypt	AUC	✓							
Wathiq-SPU	Syria	Syrian Private University	✓		✓		✓		✓	

Table 4: Teams, affiliations, and submissions by subtask and track. C and O denote Closed and Open tracks, respectively; AUC and KFUPM denote the American University in Cairo and King Fahd University of Petroleum and Minerals, respectively.

Team	Encoders	LLMs	Ensembles
Abjad AI	✓		
Al-Qahera	✓	✓	✓
AUC DL Lab	✓	✓	✓
Axiom	✓		✓
DxMultiHead	✓		✓
ENSAR	✓	✓	✓
FAAMG	✓		✓
iMak AI Lab	✓		✓
Itgan	✓		✓
Knights-ST	✓		✓
Mostafa ElAraby	✓		✓
NAMAA-Community	✓	✓	✓
NLPBeasts	✓		
Wathiq-SPU	✓		✓
Baseline	✓		

Table 5: Summary of the main modeling techniques used by the participating teams.

LLMs, including Qwen3-4B (Yang et al., 2025), Gemini-3.6-Flash (Google DeepMind, 2026), and Claude-Opus-5 (Anthropic, 2026), either as standalone predictors or for boundary refinement. Most teams used ensembles, combining architectures, random seeds, checkpoints, or configurations.

Systems also differed substantially in how they adapted models to the task. Several teams used task-adaptive or continued pretraining before supervised fine-tuning (Hossain, 2026; Sakr and Torki, 2026), while others trained jointly across multiple input variants to transfer information between segmentation conditions (Lipinski and Makarov, 2026; Almajai, 2026; ElAraby, 2026). Class imbalance was commonly addressed through weighted losses or calibrated decision thresholds (Alsudairi et al., 2026; Eid et al., 2026; Hossain, 2026; Almasri et al., 2026), and many ensemble systems tuned either model weights or classification thresholds on the development set (Mohamed and Ajlan, 2026; Ragab and Sabry, 2026; Elmasry et al., 2026; Lipinski and Makarov, 2026; Almajai, 2026). Other approaches introduced explicit structural modeling, including dependency information (Ragab and Sabry, 2026), CRF-based decoding (Elmasry et al., 2026; Salieb et al., 2026), boundary constraints (Eid et al., 2026; Almasri et al., 2026), and structured decoding for the punctuation-free settings (Ragab et al., 2026).

Open-track systems explored several ways of incorporating external data beyond the provided training data. These included pseudo-labeled Wikipedia documents (Mohamed and Ajlan, 2026), additional Arabic treebanks and corpora (Ragab and Sabry, 2026), and LLM-based refinement or pre-

diction (Mohamed and Ajlan, 2026; Emam et al., 2026; Ragab et al., 2026). Overall, submissions range from relatively simple single-encoder systems to large heterogeneous ensembles combining encoders, LLMs, and structured models.

5.3 System Descriptions

Abjad AI Alsudairi et al. (2026) use a single CAMeLBERT-MSA encoder with a linear classifier over its penultimate layer, fine-tuned using weighted cross entropy.

Al-Qahera Mohamed and Ajlan (2026) use a three-model ensemble of CAMeLBERT-MSA, AraBERTv02, and AraELECTRA. Individual checkpoints are stochastically weight-averaged, after which boundary probabilities are averaged and thresholded separately for each subtask. For the NoPnx-NP open track, they augment the training data with 2,500 pseudo-labeled Arabic Wikipedia documents. For NoPnx-PA, they instead obtain their best results by prompting Gemini-3.6-Flash.

AUC DL Lab Ragab and Sabry (2026) ensemble Arabic encoders with a syntax-aware dependency-graph model, incorporating dependency structure, and a Qwen3-4B decoder bidirectionalized using Just Pass Twice (Ewais et al., 2026) and fine-tuned with 4-bit QLoRA (Dettmers et al., 2023). Predictions are calibrated and combined using dev-tuned weights. For the open track, they add selectively sampled and leakage-filtered data from CAMeL Treebank (Habash et al., 2022), BAREC (Elmadani et al., 2025), and the Prague Arabic Dependency Treebank (PADT) (Smrž et al., 2002).

Axiom Hossain (2026) combine MARBERTv2 and CAMeLBERT-MSA in a dual-encoder ensemble. Both models undergo continued masked-language-model pretraining on the task data before fine-tuning, after which their logits are weighted and a learned decision threshold is applied.

DxMultiHead Eid et al. (2026) combine AraBERTv02, CAMeLBERT-MSA, and ARBERTv2 in a weighted ensemble trained with alignment-safe normalization, punctuation masking, and positive-class loss weighting. Logit margins are averaged across overlapping windows, followed by weighted probability fusion, a calibrated threshold, and boundary constraints.

ENSAR Emam et al. (2026) use a two-stage pipeline of an encoder ensemble followed by an LLM jury. Five encoder models spanning AraBERTv02 base and large, Arabic-BERT-large, and XLM-RoBERTa-large generate candidate boundaries, with selected models trained using input perturbation and an auxiliary contrastive objective. Predictions are averaged across overlapping windows and models, then filtered by sentence length. Two Claude-Opus-5 judges independently revise candidates using genre-specific policies, and only revisions accepted by both judges are applied.

FAAMG Elmasry et al. (2026) build an eight-model ensemble combining seed-diverse and regularized AraBERT variants with segmentation-pretrained SaT and a genre-aware BiLSTM (Hochreiter and Schmidhuber, 1997) with a conditional random field (CRF) layer (Lafferty et al., 2001) on top of CAMeLBERT. Model probabilities are fused using non-negative development-tuned weights and a global decision threshold.

iMak AI Lab Lipinski and Makarov (2026) train approximately 30 models per subtask from seven Arabic encoder families, including models jointly trained on all four input variants. Greedy forward selection retains 3-10 models according to F_1 under a calibrated decision threshold, and their probabilities are uniformly averaged.

Itgan Almajai (2026) jointly train models across all four variants using subtask-specific tags. The Closed Track system ensembles 21 models drawn from seven pretrained encoder architectures, while the Open Track uses a smaller seven-model ensemble trained on additional data. Rather than averaging predictions, the system uses L2-regularized logistic regression (Cessie and Houwelingen, 1992) to learn ensemble weights from model logits, followed by subtask-specific threshold tuning.

Knights-ST Sakr and Torki (2026) build a seven-checkpoint, cross-family ensemble for the PA closed track, combining task-adaptive pretraining with regularization (liang et al., 2021) and boundary smoothing (Zhu and Li, 2022). Per-token logits are averaged across models and overlapping windows, then thresholded to obtain predictions.

Mostafa ElAraby ElAraby (2026) use parallel punctuated and unpunctuated versions for the NoPnx-NP data. Seed ensembles of CAMeLBERT-MSA models are trained on each view, their predictions are aligned, and final boundaries are assigned using a learned set of 12 prediction patterns.

Team	PA		NP		NoPnx-PA		NoPnx-NP	
	C	O	C	O	C	O	C	O
ENSAR	**95.3** (1)	**95.3** (1)	**92.0** (1)	**92.0** (1)	90.0 (2)	90.0 (2)	**89.1** (1)	**89.1** (1)
DxMultiHead	95.0 (2)	–	91.3 (2)	–	89.5 (5)	–	86.0 (6)	–
Knights-ST	94.8 (3)	–	–	–	–	–	–	–
Al-Qahera	94.7 (4)	94.7 (2)	91.1 (4)	91.0 (3)	**91.2** (1)	**91.2** (1)	87.4 (2)	87.5 (2)
AUC DL Lab	94.5 (5)	94.6 (3)	90.8 (6)	90.9 (4)	89.2 (6)	89.7 (4)	86.0 (7)	86.0 (5)
NAMAA	94.4 (6)	94.4 (4)	91.3 (3)	91.3 (2)	89.9 (3)	89.9 (3)	87.0 (3)	87.0 (3)
Itgan	94.4 (7)	93.2 (6)	–	90.6 (5)	–	89.7 (5)	86.5 (5)	86.5 (4)
iMak AI Lab	94.1 (8)	–	91.0 (5)	–	89.5 (4)	–	85.8 (8)	–
Baseline	93.9 (9)	93.9 (5)	88.2 (9)	88.2 (6)	87.0 (8)	87.0 (6)	80.7 (10)	80.7 (7)
Wathiq-SPU	93.4 (10)	–	90.0 (7)	–	87.9 (7)	–	84.8 (9)	–
Abjad AI	91.7 (11)	–	89.7 (8)	–	–	–	–	–
NLPBeasts	89.5 (12)	–	–	–	–	–	–	–
FAAMG	–	–	–	–	–	–	86.7 (4)	–
Mostafa ElAraby	–	–	–	–	–	–	–	84.9 (6)
Axiom	–	90.2 (7)	–	–	–	–	–	–

Table 6: Participating teams' results on the Blind Test set across the Closed (C) and Open (O) tracks in terms of document-level macro $\mathbf{F_1}$. Ranks are shown in parentheses. Best results are in **bold**. The results are sorted on the PA-C, NP-C, NoPnx-PA-C, and NoPnx-NP-C columns. NAMAA denotes NAMAA-Community.

NAMAA-Community Ragab et al. (2026) combine several model families, including Arabic and multilingual pretrained encoders, bidirectionally adapted and LoRA-fine-tuned (Hu et al., 2021) LLMs, character-level SaT models, and a punctuation-restoration model. Predictions from these models are combined through weighted averaging, while a logistic maximum-entropy Markov model (McCallum et al., 2000) is used to greedily decode boundaries for the NoPnx subtasks.

NLPBeasts Salieb et al. (2026) compare two approaches for the PA Closed Track: a feature-engineered CRF and a character-word BiLSTM-CRF trained from scratch, with the latter selected as the final system. Learned word and character representations are contextualized using a two-layer BiLSTM, followed by a linear-chain CRF that decodes predictions over non-overlapping chunks.

Wathiq-SPU Almasri et al. (2026) use subtask-specific calibrated ensembles centered on CAMeL-BERT, with two-model ensembles for PA and NoPnx-PA, three independently initialized models for NP, and a context- and transfer-diverse CAMeL-BERT ensemble combined with AraELECTRA for NoPnx-NP. Predictions are combined using subtask-specific weights and thresholds, with punctuation constraints applied when available.

6 Results

Table 6 reports document-level macro F_1 on the Blind Test set for teams that submitted a system description paper. 14 teams participated in at least one subtask-track combination. Closed Track participation was higher, with 35 submissions across four subtasks, compared with 22 in the Open Track. Within the Closed Track, PA attracted the most teams (11), followed by NoPnx-NP (9), NP (8), and NoPnx-PA (7). Five teams participated in one subtask–track combination, whereas ENSAR, Al-Qahera, AUC DL Lab, and NAMAA-Community participated in all eight. 52 of the 57 team–subtask–track submissions outperformed the baseline.

6.1 Performance Across Subtasks

Performance varies substantially across the four subtasks. PA is consistently the easiest, while NoPnx-NP is the most challenging. Considering the best Closed Track result in each subtask, performance drops by 3.3 F_1 points when paragraph boundaries are removed, 4.1 points when punctuation is removed, and 6.2 points when both are removed. The same pattern holds across the five teams that participated in all four Closed Track subtasks. Relative to their PA performance, scores decrease by 3.3-5.6 points on NP, 4.1-7.4 on NoPnx-PA, and 6.2-10.5 on NoPnx-NP.

6.2 Comparison Between Teams

ENSAR ranks first in six of the eight subtask–track combinations. It leads the next-best system by 0.3, 0.7, and 1.7 F_1 points on Closed PA, NP, and NoPnx-NP, respectively, and by 0.6, 0.2, and 1.2 points in the corresponding Open settings. **Al-Qahera** leads NoPnx-PA by 1.2 points in both tracks and ranks second on Closed NoPnx-NP. Together, ENSAR, Al-Qahera, and NAMAA-Community occupy the top three ranks in five of the eight combinations.

DxMultiHead ranks second on Closed PA and ties NAMAA-Community for second on Closed NP. Among teams entering fewer settings, **Knights-ST** ranks third on Closed PA, FAAMG fourth on Closed NoPnx-NP, and Mostafa ElAraby second on Open NoPnx-NP.

Despite using different ensemble designs, AUC DL Lab, NAMAA-Community, Itgan, and iMak AI Lab achieve similar results. Their Closed Track scores span only 0.4 F_1 points on PA, 0.5 on NP, 0.7 on NoPnx-PA, and 1.2 on NoPnx-NP.

6.3 Closed and Open Tracks

Open Track participation was lower, and teams used its flexibility differently, incorporating external or pseudo-labeled data, modifying their training setup, or changing ensemble composition. Among teams reporting results in both tracks, ENSAR and NAMAA-Community obtained identical scores across all four subtasks, while differences for Al-Qahera and AUC DL Lab were at most 0.1 and 0.5 F_1 points, respectively. These differences limit conclusions about additional data, though the modest gains align with the diminishing returns reported by Elkholy et al. (2026).

6.4 Key Modeling Trends

The strongest systems consistently combine pretrained encoders with ensembling and calibration. ENSAR, Al-Qahera, DxMultiHead, NAMAA-Community, and iMak AI Lab all follow this general recipe and rank among the strongest submissions across multiple settings. Although these systems differ in ensemble size, model composition, and combination strategy, encoder ensembles emerge as the most reliable approach across the leaderboard.

LLMs provide additional benefits in some systems but are not necessary for strong performance. ENSAR and Al-Qahera achieve top results while incorporating LLM components, yet encoder-only systems such as DxMultiHead and Knights-ST remain highly competitive. Similarly, larger and more complex ensembles do not consistently outperform smaller ones, suggesting that ensemble composition and calibration are more important than scale alone.

Overall, the shared-task results indicate that strong pretrained encoders, model diversity, and careful prediction calibration are the most consistent ingredients of successful systems, while gains from LLM integration, external data, and more elaborate decoding strategies are more variable.

7 Outlook and Lessons Learned

We observed several evaluation practices that should be reconsidered in future shared tasks. Some teams used development or test data for training and calibration, while others adapted thresholds based on repeated Blind Test set leaderboard feedback. These choices partly reflect conventions from prior shared-task designs, but they reduce comparability and increase the risk of overfitting to evaluation data. Once these issues became apparent, we asked affected teams to retrain and rebuild their systems using only the data and procedures permitted by the intended evaluation setup.

We advise future shared tasks to define split usage more clearly, separate training and evaluation stages strictly, and limit Blind Test submissions.

8 Conclusion

We presented **ARASEG** 2026, the first shared task on multi-genre Arabic sentence segmentation, covering four input settings defined by punctuation availability and paragraph structure across Closed and Open Tracks. The results demonstrate the importance of both punctuation and paragraph structure. Relative to the best Closed Track result on PA, removing paragraph boundaries reduced F_1 by 3.3 points, removing punctuation by 4.1 points, and removing both by 6.2 points. The strongest systems relied primarily on diverse pretrained-encoder ensembles and careful calibration, while gains from LLMs, external data, and more elaborate architectures were less consistent. These findings establish strong benchmarks for Arabic sentence segmentation across diverse input conditions.

In future iterations, we plan to expand the shared task to additional genres and Arabic varieties, including dialectal Arabic.

Limitations

This work focuses only on segmentation for Modern Standard Arabic, and does not extend to dialectal Arabic varieties. Additionally, we restrict training to 10% of the **AraSEG** corpus. Many of the methodologies proposed by the participants may perform differently with additional training data. Finally, we use strict boundary F_1 and do not separately evaluate over- or under-segmentation or report results by genre or boundary type.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Mai Alammar, Khalil El Hindi, and Hend Al-Khalifa. 2025. English-arabic hybrid semantic text chunking based on fine-tuning bert. *Computation*, 13(6).

Ibrahim Almajai. 2026. Itgan at AraSeg shared task 2026: Diversify, don't curate – learned ensemble combination for arabic sentence segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Salam Almasri, Najat Bostaty, Bushra Al Shaabani, Raghad Alhossny, and Riad Sonbol. 2026. Wathiq-SPU at AraSeg shared task 2026: Calibrated transformer ensembles for arabic sentence segmentation across punctuation and paragraph conditions. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Abdullah M. Alshanqiti, Sami Albouq, Ahmad B. Alkhodre, Abdallah Namoun, and Emad Nabil. 2022. Employing a multilingual transformer model for segmenting unpunctuated arabic text. *Applied Sciences*, 12(20).

Mohammed Alsudairi, Taif Nono, Faris Alasmary, Anas Adel Salamah, Shouq Sadah, and Lahouari Ghouti. 2026. Abjad AI at AraSeg shared task 2026: Auxiliary loss for arabic sentence segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Touir Ameur, Mathkour Hassan, and Al-Sanea Waleed. 2008. Semantic-based segmentation of arabic texts. *Information Technology Journal*, 7.

Anthropic. 2026. Claude Opus 5 System Card.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 9–15, Marseille, France. European Language Resources Association.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2021. AraELECTRA: Pre-training text discriminators for Arabic language understanding. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 191–195, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Doug Beeferman, Adam Berger, and John Lafferty. 1999. Statistical models for text segmentation. *Machine Learning*, 34(1):177–210.

Guy Bilitski, Lev Shechter, Sonam Jamtsho, Nir Marciano, Nicola Bajetta, Rebecca Sunden, Omri Drori, Kai Golan Hashiloni, Orr Zwebner, Asaf Shina, Orna Almogi, Dorji Wangchuk, and Kfir Bar. 2026. Automatic segmentation of Classical Tibetan texts into autochthonous and allochthonous regions. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 1017–1030, Palma de Mallorca, Spain. ELRA Language Resource Association.

S. Le Cessie and J. C. Van Houwelingen. 1992. Ridge estimators in logistic regression. *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, 41(1):191–201.

Kirill Chirkunov, Younes Samih, Abed Alhakim Freihat, and Hanan Aldarmaki. 2026. Linear semantic segmentation for low-resource spoken dialects. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 34844–34861, San Diego, California, United States. Association for Computational Linguistics.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. Qlora: Efficient finetuning of quantized llms. In *Advances in Neural Information Processing Systems*, volume 36, pages 10088–10115. Curran Associates, Inc.

Muhammad Bahaa Alsayed Eid, Mohamed Khaled, and Mena Hany. 2026. DxMultiHead at AraSeg shared task 2026: Alignment-safe sequence labeling and weighted ensembling for arabic sentence segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Mostafa ElAraby. 2026. MostafaElAraby at AraSeg shared task 2026: Combining lexical and punctuation evidence across aligned text view. In *Proceedings of the Fourth Arabic Natural Language Process-*

ing Conference, Budapest, Hungary. Association for Computational Linguistics.

Mohammed Elkholy, Khalid N. Elmadani, Nizar Habash, and Bashar Alhafni. 2026. Arabic sentence segmentation across genres and punctuation conditions. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, Budapest, Hungary. Association for Computational Linguistics.

Khalid N. Elmadani, Bashar Alhafni, Hanada Taha, and Nizar Habash. 2025. BAREC shared task 2025 on Arabic readability assessment. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 239–252, Suzhou, China. Association for Computational Linguistics.

Gaser Elmasry, Abdulrahman Elbedewy, Abdelrahman Wael Atef Taha, Fahd Seddik, and Mohamed Ahmed Mohamed Abdelmoniem. 2026. FAAMG at AraSeg shared task 2026: Cross-architecture ensembling for punctuation-free arabic segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Noor Emam, Omar Mokhtar Gaber Saqr, Mostafa Gaafar, and Ali Elaswad. 2026. ENSAR at AraSeg shared task 2026: An ensemble of encoders with agentic error refinement. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Ahmed Ewais, Ahmed Hashish, and Amr Ali. 2026. Just pass twice: Efficient token classification with LLMs for zero-shot NER. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11480–11497, San Diego, California, United States. Association for Computational Linguistics.

Markus Frohmann, Igor Sterner, Ivan Vulić, Benjamin Minixhofer, and Markus Schedl. 2024. Segment any text: A universal approach for robust, efficient and adaptable sentence segmentation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 11908–11941, Miami, Florida, USA. Association for Computational Linguistics.

Google DeepMind. 2026. Gemini 3.6 Flash Model Card. `https://deepmind.google/models/model-cards/gemini-3-6-flash/`. Accessed: 2026-09-04.

Nizar Habash, Muhammed AbuOdeh, Dima Taji, Reem Faraj, Jamila El Gizuli, and Omar Kallas. 2022. Camel treebank: An open multi-genre Arabic dependency treebank. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 2672–2681, Marseille, France. European Language Resources Association.

Nizar Habash, Abdelhadi Soudi, and Tim Buckwalter. 2007. On Arabic Transliteration. In A. van den Bosch and A. Soudi, editors, *Arabic Computational Morphology: Knowledge-based and Empirical Methods*, pages 15–22. Springer, Netherlands.

Lamia Hadrich Belguith, Leila Baccour, and Mourad Ghassan. 2005. Segmentation de textes arabes basée sur l'analyse contextuelle des signes de ponctuations et de certaines particules. In *Actes de la 12ème conférence sur le Traitement Automatique des Langues Naturelles. Articles courts*, pages 451–456, Dourdan, France. ATALA.

Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long Short-Term Memory. *Neural Comput.*, 9(8):1735–1780.

Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. 2020. spacy: Industrial-strength natural language processing in python.

Md. Ajwad Hossain. 2026. Axiom at AraSeg shared task 2026: A regularized dual-model ensemble with domain-adaptive pretraining for arabic sentence segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. *Preprint*, arXiv:2106.09685.

Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. The interplay of variant, size, and task type in Arabic pre-trained language models. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 92–104, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Iskandar Keskes, Farah Benamara, and Lamia Hadrich Belguith. 2012. Clause-based discourse segmentation of Arabic texts. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC'12)*, pages 2826–2832, Istanbul, Turkey. European Language Resources Association (ELRA).

Tibor Kiss and Jan Strunk. 2006. Unsupervised multilingual sentence boundary detection. *Computational Linguistics*, 32(4):485–525.

John D. Lafferty, Andrew McCallum, and Fernando C. N. Pereira. 2001. Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data. In *International Conference on Machine Learning*, pages 282–289.

xiaobo liang, Lijun Wu, Juntao Li, Yue Wang, Qi Meng, Tao Qin, Wei Chen, Min Zhang, and Tie-Yan Liu. 2021. R-drop: Regularized dropout for neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 10890–10905. Curran Associates, Inc.

Artiom Lipinski and Ilya Makarov. 2026. iMak AI Lab at AraSeg shared task 2026: Greedy-selected arabic encoder ensembles with structural constraints for sentence segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Andrew McCallum, Dayne Freitag, and Fernando C. N. Pereira. 2000. Maximum entropy markov models for

information extraction and segmentation. In *Proceedings of the Seventeenth International Conference on Machine Learning*, ICML '00, page 591–598, San Francisco, CA, USA. Morgan Kaufmann Publishers Inc.

Asma Mekki, Inès Zribi, Mariem Ellouze, and Lamia Hadrich Belguith. 2022. Sentence boundary detection of various forms of tunisian arabic. *Language Resources and Evaluation*, 56(1):357–385.

Benjamin Minixhofer, Jonas Pfeiffer, and Ivan Vulić. 2023. Where's the point? self-supervised multilingual punctuation-agnostic sentence segmentation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7215–7235, Toronto, Canada. Association for Computational Linguistics.

Mohamed Mohamed and Hazem Ajlan. 2026. Al-Qahera at AraSeg shared task 2026: Genre-conditioned llm prompting versus supervised multi-backbone ensembling. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Lev Pevzner and Marti A. Hearst. 2002. A critique and improvement of an evaluation metric for text segmentation. *Computational Linguistics*, 28(1):19–36.

Wenjie Qiu, Yi-Chen Li, Xuqin Zhang, Tianyi Zhang, Yihang Zhang, Zongzhang Zhang, and Yang Yu. 2025. Sentence-level reward model can generalize better for aligning llm from human preference. *Preprint*, arXiv:2503.04793.

Muhammed Ragab, Khloud Al Jallad, Karim Elsayed, and Nacar Omer. 2026. NAMAA Community at AraSeg shared task 2026: Model diversity, conditional decoding, and missing punctuation in arabic sentence segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Yehia Ragab and Mohamed Sabry. 2026. AUC DL Lab at AraSeg shared task 2026: Syntax-aware graphs, bidirectionalized decoders, and encoders for arabic sentence segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Fabian Retkowski and Alexander Waibel. 2026. Paragraph segmentation revisited: Towards a standard task for structuring speech. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, pages 747–759, Palma, Mallorca, Spain. European Language Resources Association (ELRA).

Nipun Sadvilkar and Mark Neumann. 2020. PySBD: Pragmatic sentence boundary disambiguation. In *Proceedings of Second Workshop for NLP Open Source Software (NLP-OSS)*, pages 110–114, Online. Association for Computational Linguistics.

Ali Safaya, Moutasem Abdullatif, and Deniz Yuret. 2020. KUISAIL at SemEval-2020 task 12: BERT-CNN for offensive speech identification in social media. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation*, pages 2054–2059, Barcelona (online). International Committee for Computational Linguistics.

Abdelrahman Sakr and Marwan Torki. 2026. knights-ST at AraSeg shared task 2026: Cross-family ensembling with task-adaptive pretraining for arabic sentence segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Michael Reda Salieb, Andrew Aziz, Kyrollos Zakaria Soliman, and Mina Nady Serbana Shaker. 2026. NLPBeasts at AraSeg shared task 2026: A character-word bilstm-crf for paragraph-aware arabic sentence segmentation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Otakar Smrž, Jan Šnaidauf, and Petr Zemánek. 2002. Prague Dependency Treebank for Arabic: Multi-Level Annotation of Arabic Corpus. In *Proceedings of the International Symposium on Processing of Arabic*, pages 147–155, Manouba, Tunisia.

Rachel Wicks and Matt Post. 2021. A unified approach to sentence segmentation of punctuated text in many languages. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3995–4007, Online. Association for Computational Linguistics.

Zhen Xu, Sergio Escalera, Adrien Pavão, Magali Richard, Wei-Wei Tu, Quanming Yao, Huan Zhao, and Isabelle Guyon. 2022. Codabench: Flexible, easy-to-use, and reproducible meta-benchmark platform. *Patterns*, 3(7):100543.

Sane Yagi, Shehdeh Fareh, Ashraf Elnagar, Mariam Balajeed, Abdalla El-mneizel, and Mohammad Al-Badawi. 2024. Is Arabic punctuation rule-governed? *Cogent Arts & Humanities*, 11(1):2303818.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, and 41 others. 2025. Qwen3 technical report. *Preprint*, arXiv:2505.09388.

Wajdi Zaghouani and Dana Awad. 2016. Toward an Arabic punctuated corpus: Annotation guidelines and evaluation. In *The 2nd Workshop on Arabic Corpora and Processing Tools 2016 Theme: Social Media*, page 22.

Wajdi Zaghouani, Behrang Mohit, Nizar Habash, Ossama Obeid, Nadi Tomeh, Alla Rozovskaya, Noura Farra, Sarah Alkuhlani, and Kemal Oflazer. 2014. Large scale Arabic error annotation: Guidelines and framework. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation*

(LREC'14), Reykjavik, Iceland. European Language Resources Association (ELRA).

Chujie Zheng, Shixuan Liu, Mingze Li, Xiong-Hui Chen, Bowen Yu, Chang Gao, Kai Dang, Yuqiong Liu, Rui Men, An Yang, Jingren Zhou, and Junyang Lin. 2025. Group sequence policy optimization. *Preprint*, arXiv:2507.18071.

Enwei Zhu and Jinpeng Li. 2022. Boundary smoothing for named entity recognition. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7096–7108, Dublin, Ireland. Association for Computational Linguistics.